

- 1-TCP/IP is dominant protocol.
- 2- TCP/IP is a protocol used for communication with data transfer in networks, including the internet.
- 3- HTTPS is an encrypted style of HTTP, with secure communication.
- 4- SMTP (Simple Mail Transfer Protocol).
- 5- POP3 (Post Office Protocol) and IMAP (Internet Message Access Protocol).
- 6- Ethernet is a technology; Fast Ethernet and Gigabit Ethernet are standards.
- 7- A unique IP address identifies devices in a network, to get proper communication.
- 8- A subnet mask divides IP address into the network and host parts, to help routing.
- 9- A default gateway is an access point to other networks.
- 10- A DNS server translates domain names into IP addresses.
- 11- Displays IP addresses on a system.**
- 12- A MAC address is a hardware identifier unique to each network device.
- 13- DNS Round Robin distributes traffic evenly across multiple servers to balance the load.
- 14- It queries DNS servers for information about domain names.
- 15- By editing network configuration files or use tools like Netplan.
- 16- Dynamic uses DHCP to assign IPs automatically, while static is manual configuration.
- 17- YAML is readable data format, and proper indentation is critical for its structure.
- 18- Edit /etc/netplan/\*.yaml and apply the changes with netplan apply.
- 19- The server may lose network connectivity.
- 20- Netfilter.
- 21- Input, output, and forward traffic.
- 22- Use the iptables -L or ufw status commands.
- 23- Traffic is allowed by default unless explicitly blocked.
- 24- Remote SSH access to the server is denied.
- 25- Use the iptables -P command followed by the chain name and policy.

26- Saves the current iptables rules to file for later checkup.

27- UFW (Uncomplicated Firewall) provides an easy-to-use interface for managing firewall rules.

28- Displays the current status of UFW, with active rules.

29- Use command `ufw allow <profile>`.

30- Run command `ufw allow 80`.

31- Resets all UFW rules to their default state.

32- To prevent brute-force attacks by banning IPs with suspicious behavior.

33- In files located in `/etc/fail2ban/`, typically `jail.conf` or `jail.local`.

34- Use command `sudo fail2ban-client status`.

35- It logs banned IPs, failed login attempts, and related actions.